Final Design Report

Team: TrussT
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Section A4
Albro Fall 2024

——- INTRODUCTION——-

For the final truss design, we focused on increasing the max load and the load-to-cost ratio, striving to enhance our design based on what we've learned from creating the preliminary designs and by altering various factors and analyzing how those factors affect the strength and load-to-cost ratio. The goal was to create a design capable of bearing high loads relative to its cost while maintaining structural reliability. We strove to minimize uncertainty by incorporating the fit analysis equation and accounting for the uncertainty value. This analysis enabled the calculation of the uncertainty in the maximum load and ensured that potential variations in buckling strength and member forces were considered, leading to a more reliable and predictable truss performance. We did not disregard uncertainty to maximize predicted strength. Instead, the design decisions reflected a balance between addressing uncertainty and optimizing performance metrics such as maximum load and load-to-cost ratio. By factoring in uncertainty through equations and data analysis, the team ensured that the design was robust and capable of handling potential variations in material and structural behavior. The truss demonstrates efficiency and safety, reflecting a thorough understanding of the engineering principles applied to our structural design.

——- PROCEDURE——-

No changes were made to the design procedure since the preliminary design report. All MATLAB code, member numbers, and joint numbers for our final design remain the same as for our first test design, apart from slight adjustments in member lengths to allow for greater load application.

——- ANALYSIS——-

Changes in design:

The design had not been changed significantly from the preliminary designs; however, even though the overall structures are similar, there were important changes that were made to optimize the design based on what we've learned from making and analyzing the previous designs, including increasing the symmetry of the truss and utilizing the optimal height.

Calculating Max Load Uncertainty:

To account for the uncertainty in the buckling strength of each member, we used the fit analysis equation and an uncertainty value U_{fit} of ±1.54 provided by the course GSTs. We calculated a ratio based on the uncertainty derived from the data fit to determine the maximum weight our bridge can support. Given the linear relationship between the internal force of each member and the weight the bridge can withstand, there is a direct linear relationship between the uncertainty of the force in the critical member and the uncertainty in the maximum load. This relationship is expressed in Equation 1:

$$\frac{Ufit}{Fmax} = \frac{UWmax}{Wmax} \qquad \text{(Equation 1)}$$

where Fmax is the critical member force, determined using Equation 2, and Wmax is the maximum load the truss can support.

$$P_{crit} = 4863.346 * L^{-2.208} \qquad \text{(Equation 2)}$$

For the collected data based on our critical member **Fmax = 32.16 oz**, and with **Wmax = 60.58 oz**, the uncertainty in the maximum load **UWmax** is calculated as **±2.90 oz**.

—— RESULTS ——

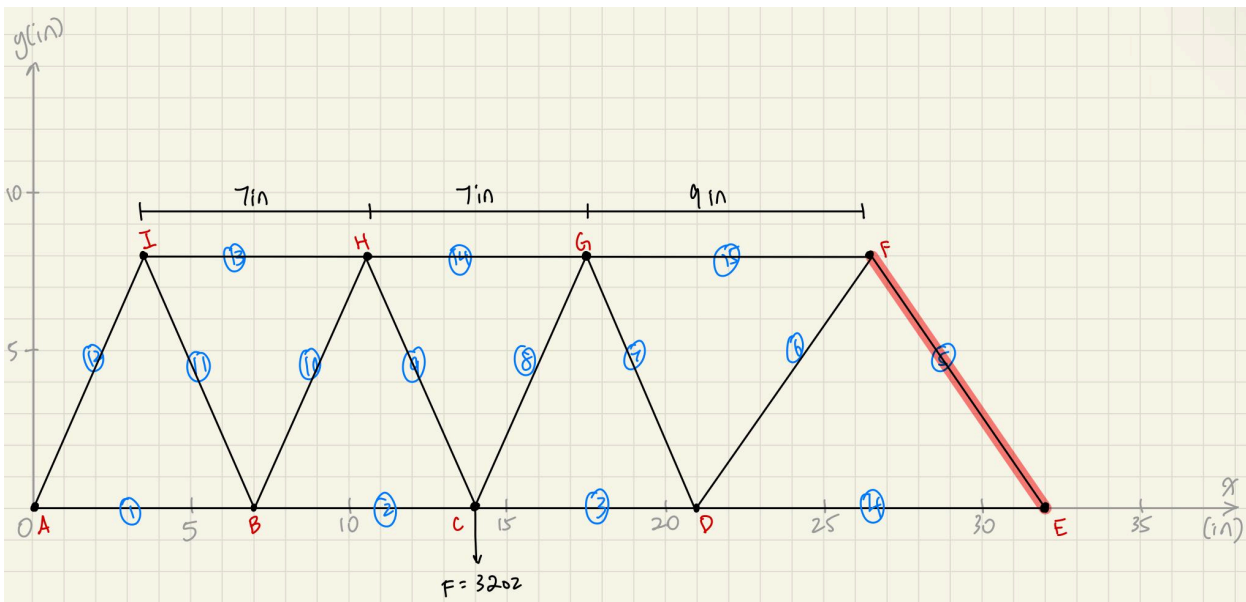

Figure 1: Diagram of our truss design with notations: member numbers, joint letters, applied force F, important distances, and the first member to buckle highlighted in red

The first truss to buckle is member 5

Member Number	Length (in)	Tension or Compression	Buckling strength and uncertainty (oz)	Magnitude of Force (oz)
1	7	T	-	7.875
2	7	T	-	23.625
3	7	T	-	25.375
4	11	T	-	9.625
5	9.708	C	32.16 ± 1.54	16.989

6	9.708	T	-	16.989
7	8.732	C	40.64 ± 1.54	15.281
8	8.732	T	-	15.281
9	8.732	T	-	19.647
10	8.732	C	40.64 ± 1.54	19.647
11	8.732	T	-	19.647
12	8.732	C	40.642 ± 1.54	19.647
13	7	C	66.22 ± 1.54	15.750
14	7	C	66.22 ± 1.54	31.500
15	9	C	38.02 ± 1.54	19.250

Table 1: Physical characteristics of all members in Truss Design 1

Summary results:

Cost of truss: $216.81
Max Load: 60.58 ± 2.90 oz
Theoretical max load/cost ratio: 0.2794 oz/$

——- DISCUSSION AND CONCLUSION ——-

Our truss design underwent various stages of development to be optimized. Initially, we designed our first truss by constructing random triangles, which resulted in a max load of 33.79 oz and a theoretical load-to-cost ratio of 0.1579 oz/$ (design 2 on the preliminary report). To make the calculation more convenient, we designed the second truss using triangles of similar lengths and angles (design 1 on the preliminary report). Thus, we discovered that triangles with more symmetry allowed for a greater max load and load-to-cost ratio (which for design 1 was 56.35 oz and 0.2505 oz/$, respectively).

From there, we tried to optimize our design by changing different variables to see which ones allowed for the best max load and load-to-cost ratio. First, we altered the height of preliminary design 1. Using the MATLAB code, it was found that the height we originally designed it with (9 in) was the optimal height for it; anything above or below that height resulted in a lower load and theoretical ratio. Then, we tried increasing the symmetry of the truss. We created a design with 6 congruent triangles from the left and 2 bigger triangles on the right where more space needed to be filled. The height for this design was set at 9 inches. The max load and ratio came out to be lower than preliminary design 1, but now understanding that there might be a more optimal height for this new truss design, we tried various heights from 7 inches to 10 inches. It was found that 8 inches had the best performance, with a max load

of 60.58 oz and a theoretical max load-to-cost ratio of 0.2794 oz/$. This is how we arrived at our optimized truss design.

We ended the project with nearly one-third of the budget remaining, showing that there's much more room for improvement in our design's max load capacity. However, time constraints limited us to minor adjustments based on the preliminary design. The reliance on a single teammate for the design also restricted creativity, thus we believe that future collaborations would benefit from group brainstorming to improve designs.

The most insightful and useful element of this project was the truss designing and analyzing process. We learned how to analyze the reaction force, moments, and such on trusses in class, but we could not have learned how to design and optimize trusses without the aforementioned project elements. It was a good challenge and learning experience trying to figure out what could make trusses stronger. This is practical knowledge that could be applied to many other real-life subjects. It also taught us to think like an engineer, constantly observing and thinking about how to make things more optimized.

The project offered two key takeaways: the construction process emphasized the gap between theory and practice while utilizing technology for calculations reduced effort and improved efficiency. Using a computer program to analyze truss designs was particularly insightful, allowing us to quickly evaluate structural integrity and test modifications. This experience enhanced our skills in programming and numerical analysis, which are vital for future engineering endeavors. One thing we would change is to have the load centered at the middle of the base so the truss could be symmetrical. It would have allowed the truss to have a greater max load and load-to-cost ratio.

——- APPENDIX A-——-

Code Output:

```
% EK301, Section A4, Trusst: Esther Xu, Doris Siu, Bryce Hwang 12/3/2024
Load: 32 oz
Member forces in oz:
m1: 7.875 (T)
m2: 23.625 (T)
m3: 25.375 (T)
m4: 9.625 (T)
m5: 16.989 (C)
m6: 16.989 (T)
m7: 15.281 (C)
m8: 15.281 (T)
m9: 19.647 (T)
m10: 19.647 (C)
m11: 19.647 (T)
m12: 19.647 (C)
m13: 15.750 (C)
m14: 31.500 (C)
m15: 19.250 (C)
Reaction forces in oz:
Sx1: -0.00
Sy1: 18.00
Sy2: 14.00
Cost of truss: $216.81
First truss to buckle is member 5
First truss to buckle length: 9.71
Predicted buckling strength: 60.58 oz
Total Length: 126.81
Max Load: 60.58 oz
Theoretical max load/cost ratio: 0.2794 oz/$
```

Code Input:

```
% Joint coordinates
X = [0, 7, 14, 21, 32, 26.5, 17.5, 10.5, 3.5];
Y = [0, 0, 0, 0, 0, 8, 8, 8, 8];

% Connection Matrix
C = [
  1,0,0,0,0,0,0,0,0,0,0,1,0,0,0;   % Joint A
  1,1,0,0,0,0,0,0,0,1,1,0,0,0,0;   % Joint B
  0,1,1,0,0,0,0,1,1,0,0,0,0,0,0;   % Joint C
  0,0,1,1,0,1,1,0,0,0,0,0,0,0,0;   % Joint D
  0,0,0,1,1,0,0,0,0,0,0,0,0,0,0;   % Joint E
  0,0,0,0,1,1,0,0,0,0,0,0,0,0,1;   % Joint F
```

```
  0,0,0,0,0,0,1,1,0,0,0,0,0,1,1;   % Joint G
  0,0,0,0,0,0,0,0,1,1,0,0,1,1,0;   % Joint H
  0,0,0,0,0,0,0,0,0,0,1,1,1,0,0;   % Joint I
];

% Reaction force matrices
Sx = [
  1, 0, 0;  % Joint A in x
  0, 0, 0;  % Joint B in x
  0, 0, 0;  % Joint C in x
  0, 0, 0;  % Joint D in x
  0, 0, 0;  % Joint E in x
  0, 0, 0;  % Joint F in x
  0, 0, 0;  % Joint G in x
  0, 0, 0;  % Joint H in x
  0, 0, 0;  % Joint I in x
];

Sy = [
  0, 1, 0;  % Joint A in y
  0, 0, 0;  % Joint B in y
  0, 0, 0;  % Joint C in y
  0, 0, 0;  % Joint D in y
  0, 0, 1;  % Joint E in y
  0, 0, 0;  % Joint F in y
  0, 0, 0;  % Joint G in y
  0, 0, 0;  % Joint H in y
  0, 0, 0;  % Joint I in y
];

% External load vector (L)
L = [0;0;0;0;0;0;0;0;0; 0;0;32;0;0;0;0;0;0;];

trusst_test(C, X, Y, Sx, Sy, L);

function trusst_test(C, X, Y, Sx, Sy, L)
[j, m] = size(C);
A = zeros(j * 2, m + 3);
A(1:j, (m + 1):(m + 3)) = Sx;
A(j + 1:j * 2, (m + 1):(m + 3)) = Sy; for jo = 1:j
for mem = 1:m
if C(jo, mem) == 1
jo2 = 0;
for jop = 1:j
if C(jop, mem) == 1 && jop ~= jo
jo2 = jop;
break;
end
end
xj = X(jo2) - X(jo);
yj = Y(jo2) - Y(jo);
d = sqrt(xj^2 + yj^2);
```

```
A(jo, mem) = xj / d;
A(jo + j, mem) = yj / d;
end
end
end
T = A\L;
tlength = 0;
length = zeros(1, m);
for mem = 1:m
jc = zeros(1, 2);
i = 1;
for jop = 1:jo
if C(jop, mem) == 1
jc(i) = jop;
i = i + 1;
end
end
xjo = X(jc(1)) - X(jc(2));
yjo = Y(jc(1)) - Y(jc(2)); length(mem) = sqrt((xjo^2) + (yjo^2)); tlength = tlength +
length(mem);
end
maxforce = 4863.346 * length.^(-2.208);
cost = tlength + 10 * j;
load = sum(L);
Tn = (T(1:m))';
R = Tn ./ load;
W = -maxforce ./ R;
Wp = W(W > 0);
Wf = min(Wp);
first2buckle = find(W == Wf, 1);
fprintf('\n%% EK301, Section A4, Trusst: Esther Xu, Doris Siu, Bryce Hwang 12/3/2024\n');
fprintf('Load: %d oz\n', load);
fprintf('Member forces in oz:\n');
for i = 1:m
torc = 'C';
if T(i) > 0
torc = 'T';
end
fprintf('m%d: %.3f (%s)\n', i, abs(T(i)), torc);
end
fprintf('Reaction forces in oz:\n');
fprintf('Sx1: %.2f\n', T(m + 1));
fprintf('Sy1: %.2f\n', T(m + 2));
fprintf('Sy2: %.2f\n', T(m + 3));
fprintf('Cost of truss: $%3.2f\n', cost);
fprintf('First truss to buckle is member %d\n', first2buckle);
fprintf('First truss to buckle length: %3.2f\n', length(first2buckle));
fprintf('Predicted buckling strength: %3.2f oz\n', Wf);
fprintf('Total Length: %3.2f\n', tlength);
fprintf('Max Load: %3.2f oz\n', Wf);
fprintf('Theoretical max load/cost ratio: %.4f oz/$\n', Wf / cost);
end
```

-——- APPENDIX B——-

Hartford Roof Collapse Discussion:

Participants:
- Chair of the Meeting: Esther Xu
- Time Recorder: Bryce Hwang
- Agenda Recorder: Doris Siu

Meeting Details:
- Date: 12/03/24
- Time: 9 pm - 9:45 pm
- Location: Mugar Library Basement

Agenda:
- Initial Thoughts
- General Summary
    - Reasons & Consequences
- Important point: Code of Ethics of Engineers
- Usage of computer programs in analysis and design
- Appropriate safety factors to be used with truss design where human life is at risk
    - Report maximum load claim for design
- Key Takeaways

Meeting Minutes:

Initial Thoughts:
- Esther: There was widespread neglect of responsibilities, no one was stepping up to address critical issues
- Doris: The focus was placed on cost-cutting and efficiency rather than safety
- Bryce: Structural concerns were ignored during construction, and I think the post-collapse evidence highlights these failures

Reasons it happened:
- Esther: The joints and members weren't properly fitted, which created instability. There was also an overreliance on flawed computer analyses that missed important factors.
- Doris: The use of substandard steel placed too much reliance on the material strength instead of ensuring construction quality.
- Bryce: The lack of a structural engineer for inspections played a significant role. The absence of redundancy in the design also led to the failure.

Consequences:
- Esther: I see the potential loss of lives and the damage to professional reputations as serious consequences.
- Doris: I think this would make it hard to regain credibility and trust for future projects.

- Bryce: The financial losses and the extended time required for repairs are significant consequences.

Important Point – Ethics:

- Fundamental Canon No. 1 from the Code of Ethics of Engineers, National Society of Professional Engineers, 2018: ”Engineers shall hold paramount the safety, health, and welfare of the public.”

Use of programs in analysis and design:

- Esther: I think the failure to account for torsional buckling in computer simulations was a critical oversight
- Doris: It’s important to anticipate discrepancies between theoretical models and real-world outcomes, and to ensure construction is accurate.
- Bryce: Precision and rigorous testing are essential before making any design decisions

Appropriate Safety Factors:

Maximum load claim for design:

- Esther: Using uncertainty values to ensure the truss can handle the maximum possible loads
- Doris: Consider every detail, like joint gaps, and to report conservative load estimations to stay on the safe side.
- Bryce: Maintain consistent precision in measurements and construction practices

Key Takeaways:

- Esther: A reminder to value feedback and adhere to professional advice.
- Doris: Sets an important standard for addressing design issues proactively rather than ignoring them.
- Bryce: Reminds us of the duty of engineers to prioritize public safety and uphold our ethical responsibilities.